

**REAL-TIME LINEARIZATION OF AN RF POWER
AMPLIFIER USING SUPPORT VECTOR REGRESSION
BASED DIGITAL PREDISTORTION TECHNIQUE**

A PROJECT REPORT

Submitted by

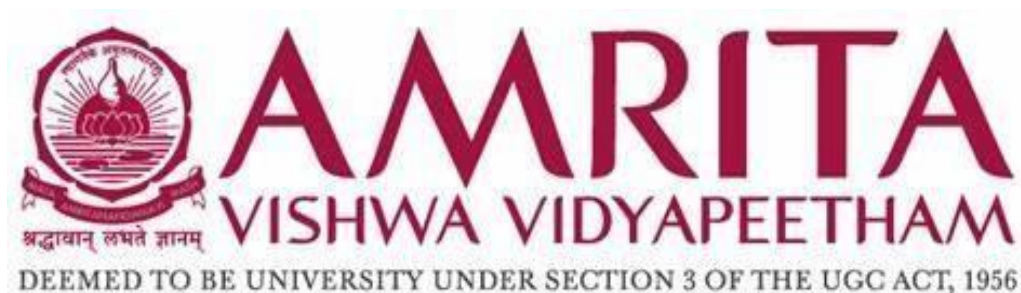
REG. NUMBER	NAME
BL.EN.U4ECE21156	Shaili Sahu
BL.EN.U4ECE21174	Susmitha GS

*in partial fulfillment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

IN

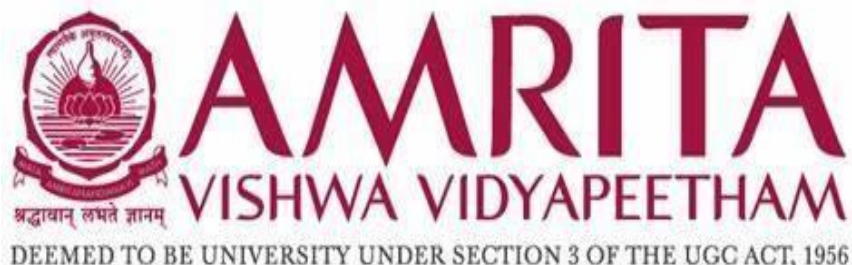
ELECTRONICS AND COMMUNICATION ENGINEERING



AMRITA SCHOOL OF ENGINEERING, BENGALURU

AMRITA VISHWA VIDYAPEETHAM

BENGALURU 560 035



BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**Real-Time Linearization of an RF Power Amplifier Using Support Vector Regression Based Digital Predistortion Technique**” submitted by

Shaili Sahu
Susmitha GS

BL.EN. U4ECE21156
BL.EN. U4ECE21174

in partial fulfillment of the requirements for the award of the **Degree Bachelor of Technology** in “**ELECTRONICS AND COMMUNICATION**” Engineering is a bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Bengaluru

Dr. R. V. Sanjika Devi

Project Guide

Assistant Professor

Dept. of ECE

ASE, Bengaluru

DATE:

Dr. T.K.Ramesh

Chairperson

Dept. of ECE

ASE, Bengaluru

DATE:

This project report was evaluated by us on

EXAMINER 1

Dr.Priya B.K
Assistant Professor
Department: ECE

EXAMINER 2

Dr. Giriraja C.V
Assistant Professor
Department: ECE

INDEX

LIST OF FIGURES.....	ii
LIST OF EQUATIONS.....	iii
LIST OF TABLE.....	iv
CHAPTER 1: INTRODUCTION.....	1
1.1 Motivation.....	1
1.2 Problem Statement.....	2
1.3 Objectives.....	3
1.4 Methodology.....	3
CHAPTER 2: STATE OF ART.....	4
2.1 Basic background.....	4
2.2 Literature Survey.....	7
CHAPTER 3: DESIGN AND DEVELOPMENT.....	19
3.1 Experiment Flow.....	19
3.2 Design and Analysis.....	19
CHAPTER 4: IMPLEMENTATION AND DEMONSTRATION.....	31
CHAPTER 5: RESULT AND DISCUSSION.....	34
CHAPTER 6: CONCLUSION AND FUTURE SCOPE.....	43
6.1 Conclusion.....	43
6.2 Future Scope.....	43
REFERENCES.....	45

LIST OF FIGURES

Figure 3.2.1: Flowchart of signal transmission through SDR	19
Figure 3.2.2: Block diagram for forward Model	22
Figure 3.2.3: Block diagram for backward Model.....	24
Figure 3.2.4: Block diagram for Linearization Model	25
Figure 3.2.5: SVR hyperplane separation line.....	28
Figure 4.1: Circuit for QPSK generation in GNU.....	31
Figure 4.2: Hardware setup using usrp board to analyse signal.....	32
Figure 4.3a: Hardware setup of RF power amplifier.	32
Figure 4.3b: Hardware setup of RF power amplifier to analyse the signal.....	33
Figure 5.1: Time domain , frequency domain, and constellation diagram for the QPSK signal in GNU software.....	34
Figure 5.2a: Tx, Rx frequency and time domain diagram transmission bandwidth 60....	35
Figure 5.2b: Tx ,Rx frequency and time domain diagram transmission bandwidth 90....	35
Figure 5.3: Amplified output of power amplifier.....	36
Figure 5.4: Plot of Input VS Output power of the rf power amplifier.....	37
Figure 5.5a: Model predicted vs actual graph for I+jQ component.....	37
Figure 5.5b: : Model predicted graph vs actual graph for I+jQ component.....	38
Figure 5.5c: DPD performance plot at gain of 7.4 v/v.....	39
Figure 5.5d: DPD performance plot at gain of 6.2 v/v.....	39
Figure 5.5e: DPD performance plot at gain of 6 v/v.....	40
Figure 5.5f : DPD performance plot at gain of 5.8 v/v	41

LIST OF EQUATIONS

Equation 3.1: Hyperplane equation.....	29
Equation 3.2: Decision rule for classification based on the hyperplane.....	29
Equation 3.3: Mathematical formulation of the optimization problem in SVM...	29
Equation 3.4: Mathematical formulation of the RBF Kernel.....	30

LIST OF TABLE

Table 5.1: Comparison table for different Gain.....	41
---	----

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported and guided us throughout the course of this university project. This work has been a valuable learning experience, and we are deeply thankful for the contributions of all those who helped bring it to completion.

First and foremost, we are extremely grateful to our project guide, Dr. Sanjika Devi, for her consistent encouragement, insightful feedback, and academic guidance. Her support throughout the project duration helped us stay focused and motivated, and her inputs were instrumental in shaping both the technical and conceptual direction of our work.

We also extend our heartfelt thanks to Dr. T. K. Ramesh, Head of the Department of Electronics and Communication Engineering at Amrita School of Engineering, Bengaluru, for providing us with this opportunity and for creating an environment that fosters learning, innovation, and research. We also wish to acknowledge the collaborative spirit and support between the two of us as teammates. The exchange of ideas, discussions, and shared responsibilities significantly contributed to the smooth execution of the project.

Last but not least, we are deeply thankful to our families for their constant motivation and emotional support, and to our peers and friends for their feedback, suggestions, and encouragement throughout this project journey.

This acknowledgment would be incomplete without recognizing the collective efforts of everyone involved. We extend our sincere appreciation to all those who played a part, directly or indirectly, in the successful completion of our project.

ABSTRACT

This project explores the use of machine learning techniques for improving the performance of radio frequency power amplifiers in communication systems. Traditional amplifiers, while essential for signal transmission, often introduce nonlinear distortions that affect signal quality. To address this, a Digital Predistortion (DPD) model was developed using Support Vector Regression (SVR) to correct the amplifier's output before transmission. Real-time signal data was collected from a hardware setup, allowing the model to be trained and tested on practical conditions. The system was evaluated through various modeling stages, and the results demonstrated clear improvements in linearity and signal fidelity. This project highlights the effectiveness of combining real-time data with intelligent modeling techniques and lays the groundwork for future advancements in RF communication systems.

CHAPTER 1 INTRODUCTION

1.1 Motivation

RF power amplifiers are key components in wireless communication systems, serving the critical function of amplifying weak signals to levels that are suitable for transmission over long distances. These amplifiers are essential in overcoming the losses caused by path attenuation, interference, and environmental factors. Without RF power amplification, signals transmitted from base stations or communication devices would be too weak to reach their destination, leading to poor signal quality and loss of data integrity. In modern wireless communication systems, such as cellular networks, satellite communication, and Wi-Fi, RF power amplifiers help maintain a robust and high-quality link by ensuring that the signal strength remains sufficiently high to cover the required transmission range, even in areas with poor signal reception.

One of the most important characteristics of RF power amplifiers is their efficiency in amplifying signals without introducing significant distortion. Ideally, an RF power amplifier should operate within its linear region, where it amplifies the input signal proportionally without introducing harmonics or nonlinearities. However, when amplifiers operate beyond their linear region, they enter a saturation state, which leads to signal distortion and inefficiencies. This problem can be exacerbated in modern communication systems that require the transmission of complex modulated signals. Techniques such as Digital Predistortion (DPD), which are part of your final year project, are employed to linearize the amplifier's response, ensuring that the amplified signal remains as close to the original as possible, even at high power levels. These techniques help extend the linear operating region of the amplifier and improve both efficiency and signal fidelity.

In addition to signal integrity, RF power amplifiers play a significant role in ensuring that wireless communication systems meet the growing demand for

higher data rates and larger coverage areas. As the demand for mobile data increases with the advent of technologies like 5G, RF power amplifiers must be capable of efficiently handling high-frequency signals while minimizing power consumption. Amplifiers must be designed to support a variety of communication standards, such as GSM, CDMA, LTE, and 5G, while maintaining optimal performance in each case. Furthermore, in highly congested frequency bands, it is crucial to minimize interference with adjacent channels. Efficient RF power amplification helps optimize spectrum usage and ensures the reliable transmission of data, even in areas with significant interference or noise.

However, despite their essential role in wireless communication, RF power amplifiers face several challenges in the current system. As communication systems evolve, particularly with the rise of 5G and the need for ever-increasing data throughput, the demand for amplifiers with higher efficiency, linearity, and power handling capabilities has increased. Traditional amplifier designs often struggle to meet these requirements, leading to issues such as high power consumption, distortion, and limited bandwidth. Furthermore, the integration of RF power amplifiers into small, portable devices presents additional design challenges, as they need to be both compact and efficient. These problems highlight the need for continued research into more advanced amplifier technologies that can deliver higher performance .

1.2 Problem Statement

The primary objective of this project is to develop a real-time Digital Predistortion (DPD) system designed to linearize the output of RF power amplifiers (PAs), thereby improving the performance and efficiency of wireless communication systems. By mitigating the nonlinearities inherent in RF power amplifiers, the DPD system aims to ensure the accurate transmission of signals, minimize power wastage, and reduce signal distortion, ultimately enhancing system reliability and spectral efficiency.

1.3 Objectives

Primary Objective: To develop a real-time digital predistortion system for linearizing an RF power amplifier, enhancing its performance and efficiency

Sub-objectives:

- Set up the SDR and VSA for signal generation and analyze the signal.
- Generate and analyze modulated signals with RF power amplifier.
- Modelling the baseband characteristic of RF power amplifier.

1.4 Methodology

- Implementation of QPSK modulated signal in SDR and analyse the signal.
- Implementation of QPSK signal and passing through power amplifier.
- Get the dataset from VSA of the amplified modulated signal.
- Modelling for the dataset obtained using DPD technique.

CHAPTER 2 STATE OF ART

This chapter reviews the current advancements in digital predistortion (DPD) techniques for RF power amplifiers, emphasizing their critical role in mitigating the nonlinearity issues inherent in these devices. RF power amplifiers are essential components in modern communication systems, tasked with amplifying signals for transmission. However, their nonlinear characteristics can lead to significant distortions that degrade signal quality and system performance. This chapter synthesizes existing research findings, categorizes various methodologies employed in DPD, and discusses their effectiveness in enhancing amplifier performance. By examining state-of-the-art literature, this chapter lays the groundwork for understanding how modern techniques can be integrated into real-time systems, ultimately contributing to improved communication reliability and efficiency.

2.1 Basic background

RF power amplifiers are crucial in communication systems, responsible for increasing the power of signals before transmission to ensure they can cover long distances and penetrate obstacles. These amplifiers are designed to operate efficiently within a specific linear range; however, when driven beyond this range, they exhibit nonlinear behaviour. This nonlinearity manifests in several detrimental ways:

Spectral Regrowth: When a signal is amplified nonlinearly, it can cause unwanted spectral components to appear outside the intended frequency band. This phenomenon leads to interference with adjacent channels, which is particularly problematic in densely packed frequency allocations common in modern wireless communication systems.

Adjacent-Channel Interference: Nonlinear amplification can result in energy spilling over into neighbouring channels, causing interference that degrades the performance of other communication systems operating nearby. This is

concerning for applications such as mobile communications and broadcasting, where multiple signals share limited bandwidth.

In-Band Distortion: Nonlinearities can introduce distortion within the signal's intended bandwidth, leading to degraded signal quality and increased error rates. This distortion adversely affects metrics like Error Vector Magnitude (EVM), which is critical for assessing the performance of modulated signals. Recent studies have highlighted these challenges extensively. For example, Borel et al. (2020) provide a comprehensive overview of how these nonlinear effects impact RF power amplifier performance and suggest various linearization techniques to mitigate them.

Digital predistortion (DPD) has emerged as a prominent technique to counteract these nonlinear effects effectively. DPD works by pre-processing the input signal before it reaches the amplifier, creating a corresponding output that compensates for the expected distortion introduced by the amplifier's nonlinear characteristics. The fundamental principle behind DPD is that by intentionally distorting the input signal in a controlled manner, it is possible to achieve a more linear output after amplification.

The implementation of DPD involves several steps:

Characterization of Nonlinearity: The first step is to accurately model the nonlinear behaviour of the amplifier using various mathematical models such as Volterra series or memory polynomial models.

Signal Processing: Once the model is established, it is used to design a pre-distorter that modifies the input signal accordingly. This process often involves complex algorithms and may utilize machine learning techniques for improved accuracy.

Real-Time Implementation: For DPD to be effective in practical applications, it must be implemented in real-time systems capable of adapting to changing conditions and maintaining performance across different operating scenarios.

Recent advancements have seen DPD techniques evolve significantly with the integration of machine learning algorithms that enhance their adaptability and efficiency. For instance, Kobal et al. (2022) propose a phase-gated recurrent

neural network model that improves the accuracy of DPD by dynamically adjusting to variations in amplifier behaviour over time.

In summary, while RF power amplifiers are essential for effective communication, their inherent nonlinearities pose significant challenges that can degrade system performance. Digital predistortion represents a viable solution to these challenges by enabling more linear amplification through sophisticated signal processing techniques.

2.2 Literature Survey

1. J. Cai, C. Yu, L. Sun, S. Chen, and J. B. King, "Dynamic Behavioral Modeling of RF Power Amplifier Based on Time-Delay Support Vector Regression," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 67, no. 2, pp. 533–542, 2019, doi: 10.1109/TMTT.2018.2884414.

This study introduces a time-delay support vector regression (SVR) technique to model the dynamic behavior of RF power amplifiers (PAs) while accounting for nonlinearities and memory effects. Key methodologies include the application of the structural risk minimization (SRM) principle, hyper parameter optimization via a grid-search technique, and the use of radial basis function kernels for accurate nonlinear behavior prediction. The experimental analysis encompassed three distinct PA types—LDMOS, single-device GaN, and Doherty GaN PAs—under various signal conditions, such as WCDMA and LTE signals. The SVR model outperformed traditional methods, including ANN-based and Volterra-series models, offering superior accuracy and computational efficiency. Additionally, the proposed model demonstrated reliable extrapolation across varying power levels, expanding its usability in real-world applications.

This inference validates the efficacy of SVR in RF PA modeling, offering a robust alternative to conventional approaches. The ability to accurately predict nonlinear behavior and memory effects across diverse PA types and input conditions makes this technique particularly suitable for modern communication systems. By enhancing digital predistortion techniques, the SVR model supports the development of efficient and scalable RF systems, particularly in scenarios demanding high precision and computational efficiency.

2. V.-K. Vu and V.-N. Tran, "Adaptive Digital Predistortion of RF Power Amplifiers Based on Memory Polynomial Model and Indirect Learning Architecture," in *2020 International Conference on Engineering and Telecommunication (EnT)*, Moscow, Russia, pp. 1-6, 2020.

This paper explores an adaptive digital predistortion (DPD) methodology to linearize RF power amplifiers (PAs) using memory polynomial models and the Indirect Learning Architecture (ILA). The approach utilizes advanced adaptation algorithms such as Recursive Least Squares (RLS), Kalman Filter (KF), and Recursive Prediction Error Method (RPEM) to estimate the DPD coefficients effectively. Simulation results demonstrate significant reduction in adjacent channel leakage ratio (ACLR), with over 25 dB improvement observed when applied to DVB-T2 signal standards. The model accurately captures nonlinear distortions and memory effects, with further optimization achieved through advanced polynomial cross-terms.

The research validates the effectiveness of the ILA in combination with adaptive algorithms for mitigating PA distortions in modern communication systems. By ensuring robust performance and computational efficiency, this work provides a scalable framework for integrating DPD techniques into RF transmitter designs, particularly in scenarios requiring precise linearization for high-bandwidth signals.

3. **T. Kobal, Y. Li, X. Wang, and A. Zhu, "Digital Predistortion of RF Power Amplifiers With Phase-Gated Recurrent Neural Networks," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 70, no. 6, pp. 3291–3300, 2022, doi: 10.1109/TMTT.2022.3161024.**

This paper introduces a novel phase-gated recurrent neural network (PG-JANET) for digital predistortion (DPD) of RF power amplifiers (PAs) under wideband excitations. The PG-JANET model incorporates signal preprocessing techniques to accurately model nonlinear interactions between amplitude and phase information, integrating these with a lightweight recurrent neural network structure. The training process includes iterative learning control for dataset preparation and sequence-to-sequence training to extract DPD parameters. Experimental evaluations compare the proposed model with other state-of-the-art methods such as GMP and LSTM-based models across various OFDM signal

bandwidths, demonstrating superior normalized mean squared error (NMSE) and adjacent channel power ratio (ACPR) performance while maintaining lower computational complexity.

This research validates the effectiveness of neural network-based models in enhancing DPD techniques for modern wideband RF systems. The PG-JANET approach highlights the integration of physical PA knowledge with machine learning, offering improved linearization accuracy and adaptability to increasing signal bandwidths, making it a promising solution for next-generation wireless communication systems.

4. Borel, V. Barzdėnas, and A. Vasjanov, "Linearization as a Solution for Power Amplifier Imperfections: A Review of Methods," in *Electronics*, vol. 10, no. 9, pp. 1073, 2021, doi: 10.3390/electronics10091073.

This review paper provides a comprehensive comparison of various linearization techniques applied to power amplifiers (PAs) to address their inherent imperfections. It categorizes these methods into three main groups: feedforward, feedback, and predistortion, examining their principles, advantages, and disadvantages. A detailed analysis of feedforward linearization highlights its ability to cancel distortion through signal and error cancellation sub-circuits, although it suffers from high power loss and sensitivity to component variations. Feedback methods, while more adaptive, face challenges at higher frequencies due to stability issues. Digital predistortion emerges as the most promising approach, leveraging adaptive algorithms to compensate for nonlinearities and memory effects effectively. The study emphasizes that each method has trade-offs in terms of spectral efficiency, power efficiency, and implementation complexity. This research underlines the need for advancements in linearization techniques to meet the stringent requirements of 5G communication systems. By presenting a comparative overview, the paper serves as a valuable resource for understanding the evolving landscape of PA linearization methods and their application in modern wireless communication systems.

5. **A. Fischer-Bühner, L. Anttila, M. D. Gomony, and M. Valkama, "Phase-Normalized Neural Network for Linearization of RF Power Amplifiers," in *IEEE Microwave and Wireless Technology Letters*, vol. 33, no. 9, pp. 1357–1360, 2023, doi: 10.1109/LMWT.2023.3290980. [5]**

The paper titled "*Phase-Normalized Neural Network for Linearization of RF Power Amplifiers*" proposes a novel phase-normalized neural network methodology to enhance the linearization of RF power amplifiers (PAs). This approach addresses nonlinear distortions in RF communications, particularly critical for 5G and future 6G technologies. The methodology incorporates a real-valued time-delay neural network (RVTDNN) with a phase normalization pre-processing step for improved modelling of nonlinear RF behaviour.

The paper identifies the limitations of traditional polynomial-based digital predistortion (DPD) models, which struggle with scalability and accuracy due to high correlations among polynomial kernels. Neural networks, particularly RVTDNNs, offer a more scalable alternative but face challenges with modelling complex baseband phases. The proposed phase-normalization technique mitigates this by aligning I/Q inputs to a reference phase, allowing the network to model memory effects effectively.

The proposed approach is experimentally validated on a Doherty GaN PA operating at 3.5 GHz, achieving an adjacent channel leakage ratio (ACLR) below -50 dBc with reduced processing complexity compared to existing models. This research demonstrates the potential of neural networks with phase normalization to significantly advance RF PA linearization for modern communication systems.

6. **I. V. Komarov and S. F. Gorgadze, "Machine learning algorithms based on variations of the least squares method for power amplifier distortion compensation," in *Proc. IEEE Systems of Signals Generating and Processing in the Field of on Board Communications (SOSG)*, 2025, pp. 1–6, doi: 10.1109/IEEECONF64229.2025.10947698.**

This paper by Igor V. Komarov and Svetlana F. Gorgadze offers an insightful comparative analysis of machine learning-based digital predistortion (DPD) techniques for power amplifier (PA) linearization. The authors emphasize the need to minimize nonlinear distortions in PAs that operate on complex, wideband signals, especially those with high peak-to-average power ratios (PAPR). These distortions, if left uncompensated, degrade the signal quality and cause spectral regrowth, leading to adjacent channel interference and regulatory violations. The study investigates two machine learning algorithms—the Least Mean Squares (LMS) algorithm and the Recursive Least Squares (RLS) algorithm—both under the indirect learning architecture framework.

The authors begin by constructing a behavioural polynomial model for the PA, which incorporates memory effects and nonlinear characteristics. Through matrix regression techniques and polynomial modelling, the system parameters are estimated using input-output signal relationships. The LMS algorithm operates using a gradient descent approach, optimizing the model coefficients by minimizing the instantaneous squared error. In contrast, the RLS algorithm updates the coefficients recursively by minimizing the cumulative squared error, weighted over time, thus adapting more rapidly to signal variations.

Simulation results indicate that the RLS algorithm outperforms LMS in terms of convergence speed and linearization quality. Specifically, the RLS algorithm reduces out-of-band emissions by approximately 8–10 dB more than LMS, and its convergence rate is nearly ten times faster. However, this performance comes at the cost of significantly higher computational complexity. The RLS method, when applied to a fifth-order polynomial model with five memory taps ($M=K=5$), demands over 70,000 multiplications per update cycle, compared to just 100 for the LMS algorithm—indicating a 700-fold increase in computational load.

Block diagrams presented in the paper illustrate the two-stage indirect learning structure training of the pre-distorter using PA input-output data and application of the trained model for signal correction. The authors also simulate spectral performance using MATLAB, showcasing the superior adjacent channel power

ratio (ACPR) performance of RLS. Nonetheless, the LMS algorithm retains a strong case for hardware implementation due to its simplicity and low resource consumption.

In conclusion, this study provides a rigorous analysis of adaptive filtering algorithms in PA distortion compensation. It underscores the trade-off between performance and complexity, showing that while RLS offers higher precision in signal linearization, its computational cost may be prohibitive for real-time embedded systems. The LMS method, despite its lower accuracy, remains a practical solution for many commercial applications.

7. **S. S. K. C. Bulusu, N. Tervo, P. Susarla, M. J. Sillanpää, O. Silvén, M. Juntti, and A. Pärssinen, "Machine learning-aided piece-wise modeling technique of power amplifier for digital predistortion," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5, doi: 10.1109/ICASSP49357.2023.10096989.**

This paper, authored by S.S. Krishna Chaitanya Bulusu and his team from the University of Oulu, proposes a novel two-stage machine learning-aided piece-wise digital predistortion (PW-DPD) technique for the behavioural modelling and linearization of power amplifiers (PAs). The motivation arises from the growing need for energy-efficient communication systems, which often push PAs to operate near saturation. However, operating in this region introduces significant nonlinear effects, including in-band distortion and spectral regrowth. Traditional approaches that rely on a single generalized memory polynomial (GMP) model struggle to adapt across the entire range of operating conditions, leading to sub-optimal performance. To address this, the authors introduce a classifier-regressor architecture that divides the input signal into classes and applies tailored linearization models to each class.

The methodology is split into two stages. In Stage I, supervised machine learning classifiers—specifically decision trees and k-nearest neighbours (KNN)—are trained using features derived from both the input signal statistics (e.g., amplitude

variance, signal energy) and the PA operating point (e.g., distance from saturation). Each sample is classified into one of five classes based on these features. In Stage II, a separate GMP model is trained for each class to compensate for nonlinearities and memory effects. The final output is reconstructed by re-aligning all class wise pre-distorted samples to form the corrected signal. This modular and scalable approach enables higher modelling accuracy with significantly lower computational complexity.

The authors validate their model using OFDM-modulated signals and a simulated nonlinear PA using a Wiener-Hammerstein structure. Their simulation framework includes analysis across various input back-off (IBO) levels, using key performance indicators like error vector magnitude (EVM) and adjacent channel leakage ratio (ACLR). The results show that the proposed ML-aided PW-DPD system outperforms a single GMP model in both EVM and ACLR, particularly at low IBO levels where distortion is more severe. At 6 dB IBO, the approach yields an EVM improvement of up to 3.3 dB and an ACLR gain of nearly 3 dB compared to the baseline model. Additionally, the classifier performance, measured using the F1 score, approaches unity across all classes when trained with five carefully selected features.

Another advantage of this method is its computational efficiency. The paper reports approximately 42% reduction in complex multiplications and additions compared to a traditional single GMP approach, making it highly suitable for hardware deployment. However, the authors acknowledge that performance can be further enhanced by optimizing the number of classes and improving class boundary selection.

8. **G. N. Chandana, G. H. Varma, J. V. Dheeraj, S. D. R. V., and D. G. Kurup, "Modeling and linearization of radio frequency power amplifiers using machine learning approaches," in *Proc. 7th IEEE International Conference on Electronics, Materials Engineering and Nano-Technology (IEMENTech)*, 2023, pp.1–6, doi:10.1109/IEMENTech60402.2023.10423509**

The paper authored by G. Navya Chandana and colleagues explores the application of kernel-based machine learning techniques—specifically Support Vector Regression (SVR) and Kernel Regression (KR)—for the modelling and linearization of radio frequency power amplifiers (RFPAs). The study targets a key challenge in wireless communications: PAs introduce nonlinear distortion and memory effects that degrade signal integrity, especially under high peak-to-average power ratio (PAPR) signals. While traditional modelling techniques like memory polynomials have been widely used, they often suffer from overfitting and lack the flexibility to generalize across various operating conditions. This work aims to evaluate the effectiveness of SVR and KR as alternatives for both behavioural modelling and digital predistortion (DPD).

The SVR algorithm, grounded in statistical learning theory, is employed to construct a nonlinear regression function that maps input signals to output values in a high-dimensional feature space using kernel functions. The authors adopt the Gaussian radial basis function (RBF) kernel due to its superior ability to capture nonlinear relationships. Unlike empirical risk minimization in neural networks, SVR employs structural risk minimization, which reduces overfitting by regularizing the model complexity. The kernel regression method used in comparison operates based on the least-squares error minimization principle but lacks the regularization robustness of SVR. Both models are implemented using the LIBSVM library, which includes automatic hyperparameter tuning via Bayesian optimization.

For experimentation, the authors simulate a Class AB GaN-based RFPA using Keysight's Advanced Design System (ADS) tool. A 10 MHz modulated baseband signal is used as input, and its distorted output is captured for analysis. The modelling accuracy is assessed using the Normalized Mean Square Error (NMSE), and the in-band linearization is measured post-DPD. Results show that while memory polynomial models outperform SVR and KR in forward modelling (i.e., PA characterization), SVR achieves superior DPD performance. Specifically,

SVR-based DPD achieves an NMSE of -56.2 dB, significantly outperforming kernel regression (-44.2 dB) and polynomial-based DPD (-42.7 dB).

These findings suggest that SVR not only generalizes better but also mitigates the overshoot issues seen with polynomial models, particularly in regions of strong nonlinearity. Although polynomial models can be simpler and faster to compute, they lack the adaptability and regularization features of SVR. Moreover, KR offers an intermediate performance between the two extremes.

9. M. Lin, M. Li, and D. Liu, "PD-type iterative learning control for support-vector-machine-based digital pre-distortion of power amplifier," in *Proc. IEEE 5th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC)*, 2021, pp. 1647–1652, doi: 10.1109/IAEAC50856.2021.9390894.

This paper presents an innovative hybrid approach combining Proportional-Derivative Iterative Learning Control (PD-ILC) and Twin Support Vector Regression (TSVR) for digital predistortion (DPD) of power amplifiers (PAs). Authored by Meng Lin et al., the study addresses the challenge of nonlinear distortion in PAs—especially in scenarios involving wideband, high-throughput communication signals. Traditional DPD methods heavily depend on accurate modelling of the amplifier behaviour, but these techniques often struggle with dynamic nonlinearities and memory effects. To overcome this, the paper introduces a two-stage process where the PD-ILC algorithm is first used to iteratively determine the optimal input signal that drives the PA toward a linear response. Once this signal is identified, it becomes the training target for the TSVR-based pre-distortion model.

The PD-ILC operates by iteratively minimizing the error between the desired output and actual output of the PA using a dynamic gain-based learning matrix. The key enhancement over conventional ILC methods is the incorporation of both proportional and differential terms in the error feedback, which significantly improves convergence speed and tracking performance.

Mathematical analysis confirms that this PD-type formulation guarantees monotonic convergence of the error norm. This iterative optimization removes the reliance on explicit PA modelling, which is often difficult or inaccurate in practical settings.

In the second stage, a TSVR model is trained using the optimal input signal from PD-ILC as the target output. TSVR, a machine learning variant of support vector machines, is effective in modelling nonlinear systems with high generalization and robustness. The regression function is constructed using Gaussian kernels, and the optimization process is formulated with dual objective functions to maintain bounded model complexity and fit. Simulation and experimental results show that the proposed PDILC-TSVR model achieves superior performance in both normalized mean square error (NMSE) and adjacent channel power ratio (ACPR), compared to traditional methods such as memory polynomial (MP) models or direct inverse modelling.

To validate the approach, the authors conducted RF measurements using a GaN Class-F PA at 2.1 GHz with 20 MHz LTE signals. The PDILC-based TSVR-DPD achieved an ACPR of -54.78 dB, outperforming the inverse modelling structure (-50.68 dB) and PDILC with MP model (-52.64 dB). Additionally, the AM/AM and AM/PM plots confirm the effective linearization of gain and phase distortion. These results demonstrate that integrating iterative control theory with machine learning offers a powerful solution for adaptive DPD design, especially when amplifier characteristics are complex and nonlinear. The hybrid methodology also proves to be hardware-friendly due to its fast convergence and scalable architecture.

- 10. A. Fischer-Bühner, L. Anttila, A. Brihuega, M. D. Gomony, and M. Valkama, "Predistortion of GaN power amplifier transient responses in time-division duplex using machine learning," *IEEE Microwave and Wireless Technology Letters*, early access, Apr. 2025, doi: 10.1109/LMWT.2025.356122**

This paper address a critical challenge in time-division duplex (TDD) systems using gallium nitride (GaN) power amplifiers (PAs)—namely, transient gain distortion when transitioning from idle to active transmission. This effect is particularly problematic in 5G and emerging 6G systems, where PAs frequently alternate between uplink and downlink operation. The transient response, typically lasting from microseconds to milliseconds, affects the first few symbols of each downlink slot, causing signal degradation that traditional digital predistortion (DPD) techniques cannot mitigate due to their short-term memory window. To resolve this, the authors propose a novel machine learning-based predistortion framework that combines a recurrent transient gain model with an existing short-term neural network (NN)-based DPD.

The methodology involves two key components. First, the transient gain behaviour is modelled using a low-rate recurrent structure built upon infinite impulse response (IIR) filters. These filters are configured to capture both first-order (decaying) and second-order (oscillatory) system responses, which are commonly observed in GaN PAs due to thermal effects, electron trapping, and power supply instability. The IIR filters are initialized with structured coefficients to avoid gradient explosion or vanishing during training and are optimized using backpropagation through time (BPTT). To reduce computational burden, the model is trained at a sampling rate 512 times lower than the DPD rate, and the results are upsampled to match the actual baseband rate.

In the second stage, the transient gain model is integrated with the short-term DPD using three possible configurations: (A) gain correction after DPD, (B) gain correction before DPD, and (C) augmentation of the DPD with gain model input. Configuration C, where the gain prediction is fed into the NN DPD as an additional input, proves to be the most effective. This structure allows the DPD to learn the joint behaviour of short-term and transient effects, thereby achieving seamless compensation. The authors validate this method using a 3.6 GHz GaN PA with LTE-like 100 MHz TDD waveforms.

Measured results show that configuration C significantly reduces symbol-level error vector magnitude (EVM) in the initial OFDM symbols of each slot, outperforming the other two configurations and traditional DPD baselines.

This work is a breakthrough in addressing long-term memory effects in RF PAs through ML. It demonstrates that combining transient compensation with real-time DPD not only enhances signal quality but also ensures better compliance with spectral regulations. The approach is scalable, data-driven, and adaptive, making it suitable for future TDD-based systems operating in mm wave and massive MIMO environments.

CHAPTER 3

DESIGN AND DEVELOPMENT

3.1 Experiment Flow

- Generation of QPSK modulated with different Bandwidths in GNU Radio
- Transmitted the modulated signal to the SDR
- Analyzed the signal using VSA from SDR
- Transmitted the modulated signal to a real-time power amplifier via a VSG
- Acquired real-time dataset of the amplified modulated signal from the power amplifier via the VSA for modelling of the amplifier.
- Trained SVR models with RBF kernel for forward and backward modeling to build the PA linearization system

3.2 Design and Analysis

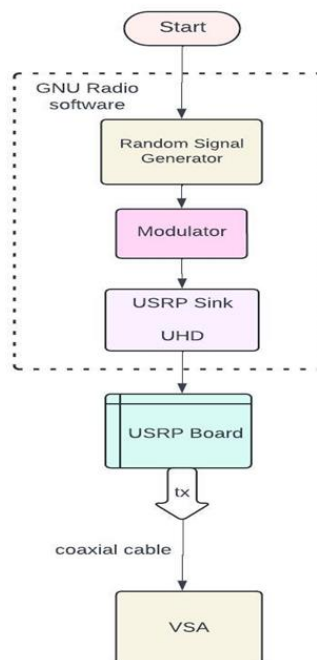


Figure 3.2.1: Flowchart of signal transmission through SDR

QPSK Modulation Using GNU Radio

The project began with the implementation of QPSK modulation using GNU Radio. GNU Radio was used to simulate a QPSK-modulated signal in software as shown in fig 3.2.1,

providing an ideal representation of the signal's behaviour in terms of bandwidth and modulation characteristics. This software-generated signal served as the input for further hardware-based analysis and testing.

Signal Transmission Through SDR

The simulated QPSK signal was transmitted using the USRP B210 board, a Software-Defined Radio (SDR) with an inbuilt power amplifier. The USRP B210 allowed for the direct transmission of the QPSK signal, which was then analyzed using a Vector Signal Analyzer .

Transition to RF Power Amplifier

To enhance the signal transmission and reduce the limitations observed in the SDR setup, the project transitioned to using an external EVAL-CN0417-EBZ 2.4 GHz RF Power Amplifier. This external amplifier was integrated into the signal path to provide better performance and reduce signal degradation.

1. Signal Preparation:

- The modulated signal generated in GNU Radio was converted into In-phase (I) and Quadrature (Q) components. This conversion created a dataset suitable for upload to the Vector Signal Generator (VSG).

2. Signal Path Setup:

- The I and Q datasets were uploaded to the VSG, which transmitted the signal to the external RF amplifier.

- A 30 dB attenuator and a coaxial cable were connected at the amplifier's output to manage the signal strength before passing it to the VSA for analysis and the output was captured.

Identifying the Nonlinear Region

The amplified signal was tested across a range of frequencies to determine the nonlinear region of the RF amplifier. The input-output characteristics of the amplifier were measured and plotted to identify the point where the amplifier transitioned from a linear to a nonlinear response.

Delay Compensation

To ensure accurate modelling of the amplifier's behaviour, cross-correlation analysis was performed to identify and compensate for delay components introduced by the system. The delay-adjusted dataset allowed for precise alignment between the input and output signals, which was critical for modelling. Finally, the adjusted dataset of I and Q components was used to model the nonlinear behaviour of the amplifier. Support Vector Regression (SVR) with a RBF kernel was employed for this purpose.

Modelling methodology

Modelling helps in understanding the PA's behaviour and designing appropriate correction mechanisms. In our project we have followed forward modelling, backward modelling, and finally integrating them into a complete DPD-PA Linearisation model.

Forward Modelling

Forward modelling is a critical step in the development of a digital predistortion (DPD) system. It involves constructing a mathematical or machine learning-based model that accurately simulates the real-world behaviour of a Radio Frequency (RF) Power Amplifier (PA). The primary goal of forward modelling is to learn

how the amplifier processes input signals and produces corresponding distorted outputs, taking into account both nonlinearities and memory effects.

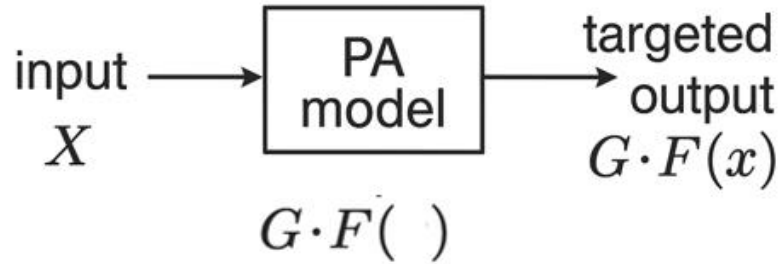


Figure 3.2.2: Block diagram for forward Model

As illustrated in Fig 3.2.2, the modelling process begins with the input signal X , which represents the time-domain baseband I/Q signal. This input is passed through the PA model, which is trained to reproduce the amplifier's response. The output of the model is denoted as $G \cdot F(x)$, where $F(x)$ represents the nonlinear transformation imposed by the PA, and G represents the fixed gain applied to the signal. This simulated output is used to validate and evaluate the effectiveness of the predistortion model. By learning the amplifier's behaviour, the forward model serves as a digital replica of the PA, enabling offline testing and verification of DPD performance.

To capture the amplifier's complex nonlinear characteristics, Support Vector Regression (SVR) was chosen as the modelling approach. SVR is a supervised learning method well-suited for regression tasks involving nonlinear and high-dimensional data. To enhance the model's ability to represent nonlinear behaviour, the Radial Basis Function (RBF) kernel was used. This kernel function maps the input features into a higher-dimensional space, enabling the SVR to fit nonlinear relationships without explicitly defining complex basis functions. The RBF kernel was selected after empirical comparisons with other kernel types such as linear, polynomial, and spline kernels. Among these, the RBF kernel consistently delivered superior performance in terms of Normalized Mean Squared Error (NMSE), making it the optimal choice for this task.

The training of the forward model was conducted using a memory-based input structure. A memory depth of 4 was selected to account for memory effects inherent in the PA. This means that each input to the model consisted of the current and three previous samples, resulting in a 10-dimensional input vector (5 I and 5 Q values). The model was trained separately for I and Q outputs using SVR, and standard scaler was applied to normalize both input and output data for numerical stability.

A total of 20,000 samples were used for training the model, and another 20,000 samples were reserved for testing. This separation ensured that the model was evaluated on unseen data, allowing assessment of its generalization ability. The trained forward model is essential in evaluating the overall DPD system, as it simulates how the actual PA would respond to pre distorted signals.

The SVR hyperparameters were fine-tuned as follows:

- C (Regularization Parameter): Controls the trade-off between achieving a low error on the training data and maintaining a smooth prediction function. Higher values of C allow the model to focus on reducing training errors but may lead to overfitting.
- ϵ (Epsilon Insensitive Loss): Defines a margin within which no penalty is given for errors. It determines the model's tolerance to small errors.
- γ (Gamma): Determines the influence range of individual training points. A small γ value implies a wider influence, while a large γ value makes the model focus only on nearby points.

Grid search and trial-error experiments were conducted to identify the optimal combination of these parameters .

Backward Modelling

Backward modelling plays a vital role in the implementation of Digital Predistortion (DPD) systems for linearizing Power Amplifiers (PAs). It refers to the process of learning an inverse function of the PA's behaviour, such that the

system can generate a pre-distorted input signal that, after passing through the actual nonlinear PA, results in a linear output. This modelling strategy is particularly useful because directly inverting a nonlinear and memory-affected system is mathematically complex and often infeasible. Instead, machine learning methods like Support Vector Regression (SVR) offer a data-driven solution to approximate the inverse behaviour of the PA.

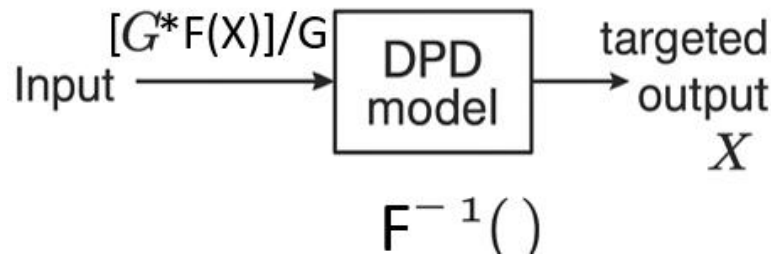


Figure 3.2.3: Block diagram for Backward Model

The working principle of this approach is illustrated in Fig. 3.2.3. In this diagram, the input to the DPD model is the measured output of the PA, normalized by a constant gain factor G . The normalization, shown as $G \cdot F(X)/G$, ensures scale consistency and allows the model to focus on learning only the nonlinear distortions without being affected by constant amplitude scaling. The target output of the DPD model is the original input signal X , which was applied to the PA during data collection. The DPD model thus approximates the function, i.e., the inverse of the PA's nonlinear function. When this DPD-generated signal is passed back through the real PA, it counteracts the amplifier's distortions and ideally results in a linear and clean output.

For training the backward model, the same memory-based architecture used in the forward model was adopted. A memory depth of 4 was selected, meaning that for every prediction, the model used the current and four previous samples of both the in-phase (I) and quadrature (Q) components. This results in a 10-dimensional input feature vector that captures the time-dependent nature (memory effects) of the PA. Separate SVR models were trained for the I and Q components of the signal. The Radial Basis Function (RBF) kernel was chosen after testing various

kernel options like linear, polynomial, and spline kernels. The RBF kernel consistently yielded better performance in terms of Normalized Mean Squared Error (NMSE) and signal fidelity.

The training process involved a dataset of 20,000 samples, using the normalized PA output as the input and the original signal as the target. Input features and targets were scaled using standard scaler to ensure better convergence during SVR training. Once trained, this backward model effectively becomes the pre-distorter. When applied to new data, it generates a modified input that cancels out the PA's nonlinear behaviour, ensuring that the PA's output is as close to linear as possible.

Linearization Model

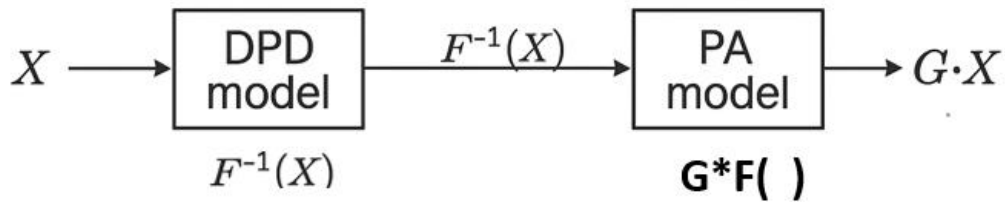


Figure 3.2.4: Block diagram for Linearization Model

The linearization model as shown in Fig. 3.2.4 involves the backward model (DPD) takes the distorted PA output and predicts the ideal pre-distorted input signal. This predicted pre-distorted signal is then passed through the PA forward model, which simulates the PA's behaviour. The output of the forward model is expected to resemble the original input signal, indicating linearization. The linearization model was implemented by integrating the trained Digital Predistortion (DPD) backward model with the forward PA model in a sequential manner, creating a digital simulation loop for evaluating the linearization performance. The process began by loading the real measured I and Q components from the collected dataset, which contained the input and output signals of the RF power amplifier. The backward model, which had been trained using Support Vector Regression (SVR) with an RBF kernel, was responsible for

generating the pre-distorted signal and was loaded to the current model ,similarly the forward model was also imported and the input was given to the backward model and the output of the backward model was given as the input for the forward model and the output was a linearized output

Programming Language: Python

Python was selected as the primary programming language for this project due to its simplicity, readability, and rich ecosystem of libraries and frameworks tailored for machine learning and data science. Its flexibility and versatility make it an ideal choice for implementing complex algorithms and data analysis tasks.

Libraries and Frameworks

- **Pandas:** This library is essential for data manipulation and analysis. It provides efficient data structures like Data Frames and Series, enabling seamless data loading, cleaning, and transformation.
- **NumPy:** As the fundamental library for numerical computations in Python, NumPy is used for handling arrays, matrices, and mathematical operations. It forms the backbone of many scientific computing and machine learning applications.
- **Scikit-learn:** This powerful machine learning library offers a wide range of algorithms, including Support Vector Regression (SVR), for model building, training, and evaluation. It also provides tools for data preprocessing, model selection, and hyperparameter tuning.
- **Matplotlib:** This versatile plotting library is employed to visualize the performance of the SVR model. It enables creating various types of plots, including scatter plots, line plots, and bar charts, to analyze the relationship between input and output signals.

Development Environment: Google Colab

Google Colab, a cloud-based Jupyter notebook environment, was utilized for its ease of use and accessibility. It provides a seamless platform for writing and executing Python code, as well as for training and deploying machine learning models.

Data Source

For the project, real-world signal data was collected from a physical Radio Frequency Power Amplifier (RFPA) setup to ensure that the modeling and digital predistortion (DPD) techniques were evaluated on practical, non-simulated conditions. A signal was generated and transmitted through the power amplifier using a Vector Signal Generator (VSG). The output of the amplifier, which included the distorted I (in-phase) and Q (quadrature) components, was captured using a Vector Signal Analyzer (VSA). This setup allowed us to gather both the input signals fed into the PA and the corresponding output signals generated by the PA, thereby capturing the complete input-output behavior of the system under real operating conditions.. This file contains raw data pertaining to input and output signal components, specifically:

- input_icomp: In-phase component of the input signal
- input_qcomp: Quadrature component of the input signal
- output_icomp: In-phase component of the output signal
- output_qcomp: Quadrature component of the output signal

To prepare the training dataset for modeling, a Python script was developed to extract and format signal component data into a structured Excel file. The input data was sourced from four separate text files containing the in-phase (I) and quadrature (Q) components of both the input and output signals of a power amplifier. Specifically, the script reads the second numeric value from each line in these text file representing the I/Q signal components and stores the first 20 values from each file into separate arrays. To ensure consistency across all arrays, each is padded with zeros if it contains fewer than 20 values. These arrays are then combined into a 2D array structure representing: input I, input Q, output I, and

output Q components. The combined dataset is exported into a formatted Excel sheet using the openpyxl library, resulting in a file titled combined_data.xlsx. This formatted dataset was subsequently used as input for training the machine learning models employed in the forward and backward modeling of the power amplifier.

Machine Learning Model: Support Vector Regression (SVR)

Support Vector Regressor (SVR) is a supervised machine learning algorithm. It is specifically used for regression tasks, making predictions about continuous target variables. It is derived from the Support Vector Machine (SVM) algorithm, which is commonly used for classification. The primary goal of SVR is to find a function that deviates from the actual target values by a margin of tolerance (ϵ) while minimizing model complexity.

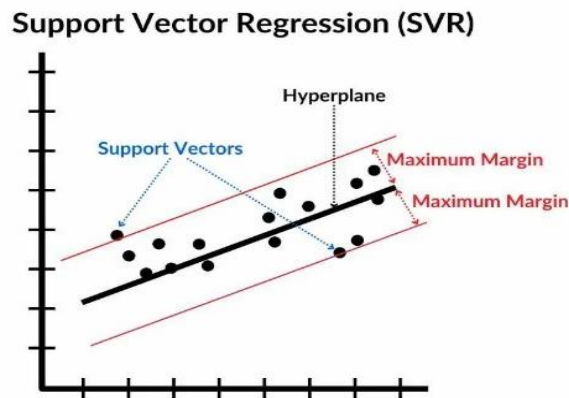


Figure 3.2.5: SVR hyperplane separation line

SVR Algorithm Steps:

1. **Map Data:** Transform data into a higher-dimensional space using kernel functions to handle complex relationships.
2. **Find Optimal Hyperplane:** Identifies the hyperplane as shown in Fig.3.2.5 that maximizes the margin, minimizing classification errors.
3. **Classify New Data:** Assign new data points to a class based on their position relative to the hyperplane.

- **Basic Formula:**

The equation for the linear hyperplane in the figure 3.2.5 can be written as:

$$w^T x + b = 0 \quad (3.1)$$

Equation : Hyperplane equation.

The vector W represents the normal vector to the hyperplane. i.e the direction perpendicular to the hyperplane. The parameter b in equation 3.1 represents the offset or distance of the hyperplane from the origin along the normal vector w .

Linear SVR classifier :

$$\hat{y} = \begin{cases} 1 & : w^T x + b \geq 0 \\ 0 & : w^T x + b < 0 \end{cases} \quad (3.2)$$

Equation: Decision rule for classification based on the hyperplane.

- **Optimization :**

For Hard margin linear SVR classifier

$$\begin{aligned} & \underset{w,b}{\text{minimize}} \frac{1}{2} w^T w = \underset{W,b}{\text{minimize}} \frac{1}{2} \|w\|^2 \\ & \text{subject to } y_i(w^T x_i + b) \geq 1 \text{ for } i = 1, 2, 3, \dots, m \end{aligned} \quad (3.3)$$

Equation : Mathematical formulation of the optimization problem in SVM.

target variable or label for the i th training instance is denoted by the symbol t_i in this statement. And $t_i = -1$ for negative occurrences (when $y_i = 0$) and $t_i = 1$ positive instances (when $y_i = 1$) respectively.

Hyperparameter Tuning

Importance of Hyperparameter Tuning

Hyperparameters are settings that control the learning process of a machine learning model, such as SVR. They are not learned from the training data but are set before training begins. Selecting appropriate hyperparameters is crucial for

achieving optimal performance, controls the trade-off between fitting the data and maintaining generalization and helps avoid overfitting or underfitting.

Radial Basis Function kernel

The Radial Basis Function kernel was employed as the core kernel function within the Support Vector Regression framework to model the nonlinear behaviour of the RF power amplifier and to develop the digital predistortion model. The RBF kernel is well-suited for such tasks due to its ability to handle complex nonlinear relationships by transforming the input space into a higher-dimensional feature space where linear regression can be effectively performed

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (3.4)$$

Equation : Mathematical formulation of the RBF Kernel

- x_i and x_j are input feature vectors,
- $\|x_i - x_j\|^2$ is the squared Euclidean distance between the two feature vectors,
- Γ gamma is the kernel coefficient that defines the influence of individual training samples. A larger γ results in a narrower influence range, while a smaller γ allows a broader influence.

The key advantage of the RBF kernel in this project is its local response property, where the similarity between data points decays exponentially with distance. This property enables the SVR model to effectively capture both nonlinearities and memory effects in the amplifier's behavior without the need for explicitly defining polynomial or higher-order terms.

CHAPTER 4

IMPLEMENTATION AND DEMONSTRATION

This chapter explains the steps taken to design, implement, and showcase the system, including the tools and techniques used. The chapter also describes how different parts, like hardware and software, were put together to create a complete and functional solution

4.1 Demonstration of Hardware Setup:

QPSK Implementation Using GNU Radio:

The implementation began by using GNU Radio to simulate a QPSK modulated signal. This software simulation provided an ideal representation of how the modulated signal would behave, including bandwidth characteristics and circuit is shown in figure 4.1.

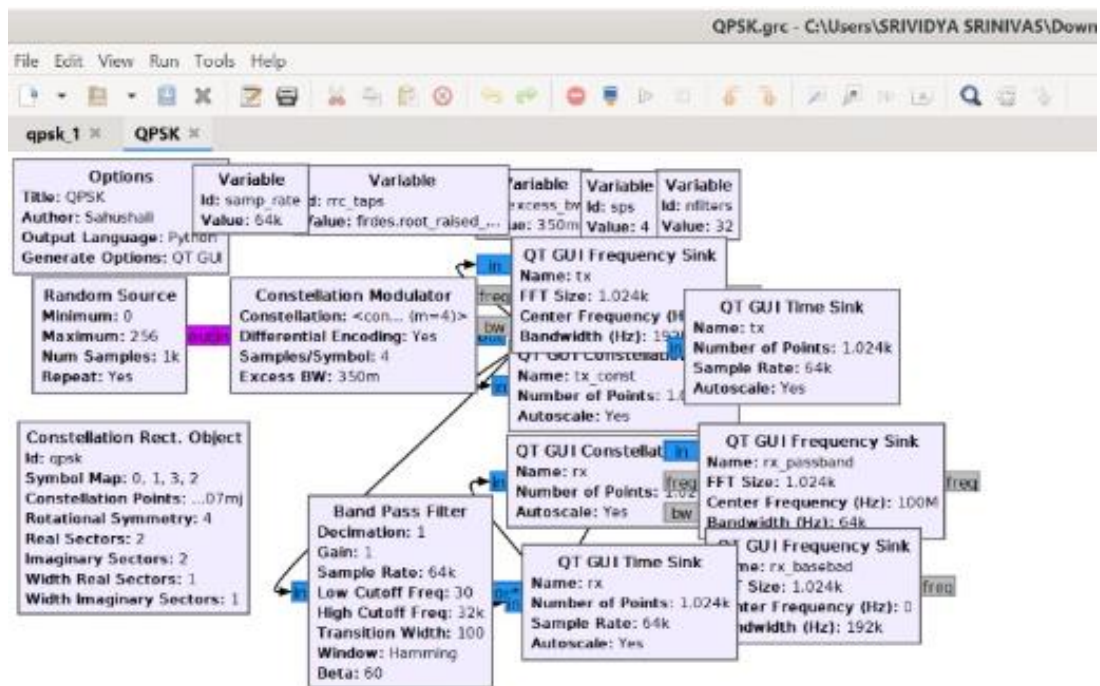


Figure 4.1: Circuit for QPSK generation in GNU

The simulated signal was then transmitted to the Software-Defined Radio (SDR) system. The SDR used was the USRP B200 board, which has an inbuilt power amplifier. The transmitted signal was analyzed using a Vector Signal Analyzer as shown in figure 4.2 .

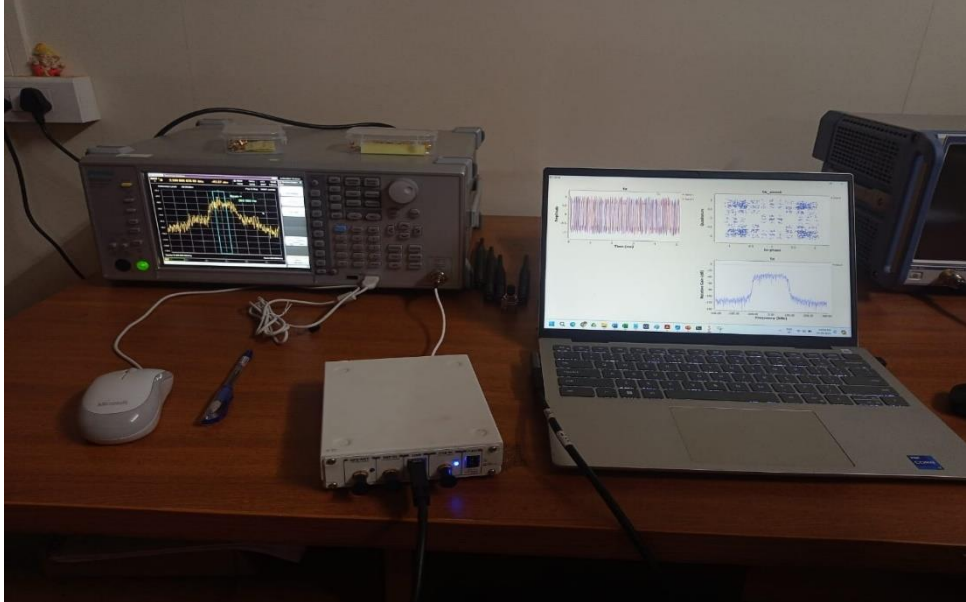


Figure 4.2: Hardware setup using usrp board to analyse the signal

To better understand the behavior of the signal and address the observed degradation, we utilized the SDR to study and analyze our QPSK-modulated signal in detail. This allowed us to evaluate the impact of the SDR's internal components, such as coaxial cables and the inbuilt amplifier, on signal quality. Based on these observations, we transitioned to using an external EVAL-CN0417-EBZ 2.4 GHz RF Power Amplifier to ensure more accurate testing and analysis.

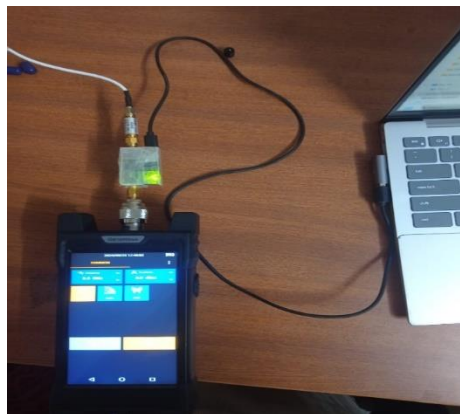


Figure 4.3.a: Hardware setup of RF power amplifier

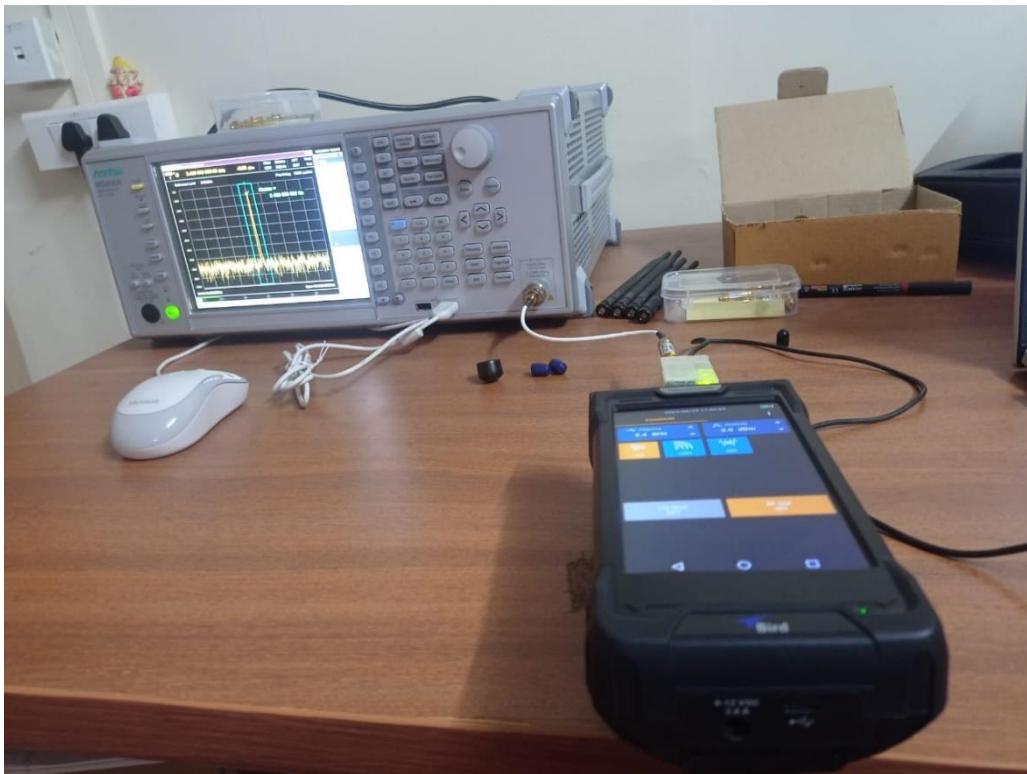


Figure 4.3.b: Hardware setup of RF power amplifier to analyze the signal

Readings were taken across various frequencies to determine the point at which the amplifier begins to saturate. This analysis helped identify the nonlinear region of the amplifier's operation.

The SVR regression effectively is captured the nonlinear characteristics of the amplifier, providing a robust and reliable model for further analysis and implementation of Digital Predistortion (DPD).

CHAPTER 5

RESULT AND DISCUSSION

This chapter focuses on the outcomes of the experiments performed during the project and explains their significance. The results are compared with the expected behavior to understand how well the system achieves its objectives.

5.1 Evaluation and Results:

QPSK Implementation Using GNU Radio

- The QPSK signal simulated using GNU Radio produced a clean modulated signal with defined bandwidth and constellation points.
- The signal characteristics were observed in the frequency and time domains, confirming proper modulation as shown in figure 5.1.

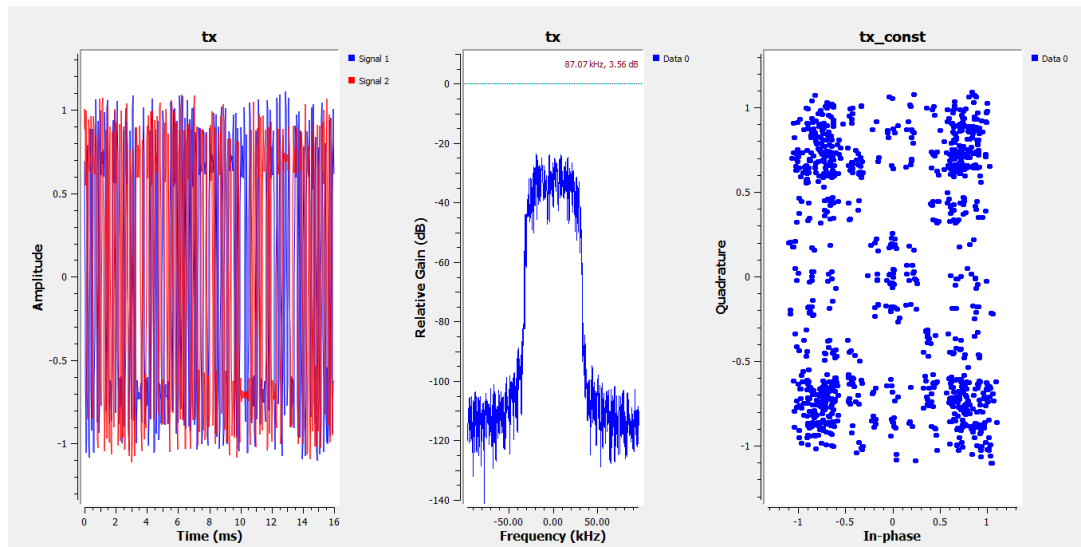


Figure 5.1: Time domain ,frequency domain, and constellation diagram for the QPSK signal in GNU software

SDR's Power Amplifier

The QPSK-modulated signal transmitted via the SDR's inbuilt power amplifier was observed using the **VSA**. The frequency-domain analysis revealed significant degradation in both power and bandwidth.

Observation:

- The bandwidth of the output signal was reduced to half compared to the ideal signal.
- Power loss was evident across all tested bandwidth settings, indicating limitations in the SDR's inbuilt power amplifier.
- As the bandwidth increases the non linearity increases which is shown in figure 5.2.a and 5.2.b respectively.

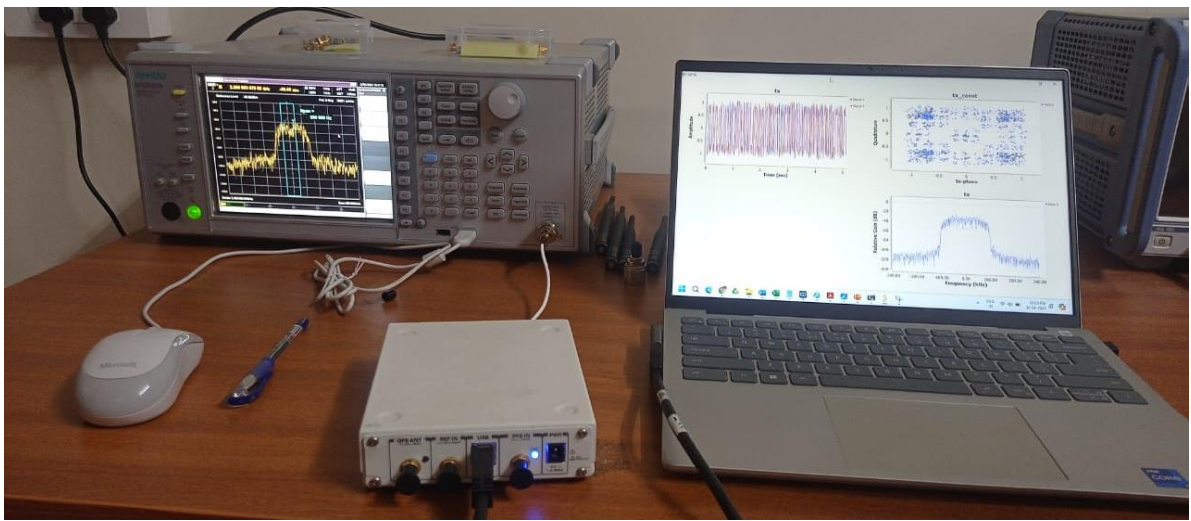


Figure 5.2.a. Tx , Rx frequency and time domain diagram transmission bandwidth 60

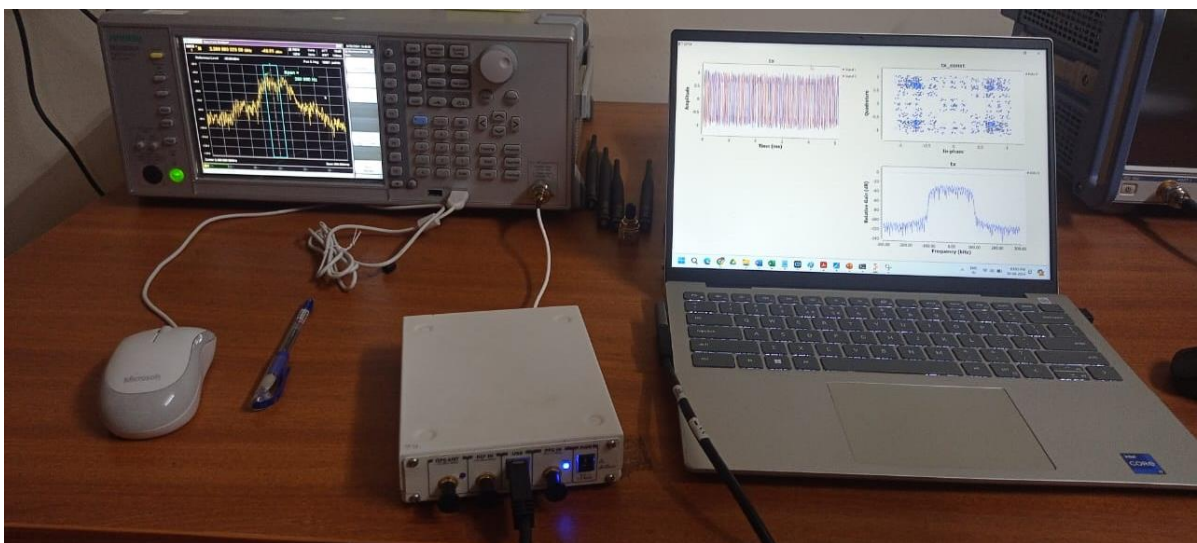


Figure 5.2.b. Tx , Rx frequency and time domain diagram transmission bandwidth 90

Implementation using RF Power Amplifier

The QPSK-modulated signal was processed through the EVAL-CN0417-EBZ 2.4 GHz RF Power Amplifier. The output signal was observed via the VSA, showing significant improvements in power and bandwidth retention compared to the SDR's inbuilt amplifier.

Observation:

- The external amplifier preserved the signal's bandwidth and maintained higher output power.
- Minimal degradation was observed in signal quality ,power as shown in fig 5.3 .



Figure 5.3: Amplified output of power amplifier

Identifying the Nonlinear Region

The amplifier's behaviour was tested across different frequencies, and readings were recorded to plot the output characteristics. The graph in figure 5.4 revealed the frequency range where the amplifier transitioned into its nonlinear region.

Observation:

- The amplifier exhibited linear behavior at lower frequencies, with minimal distortion. Nonlinear behavior was observed beyond a certain frequency threshold, where saturation effects became evident.

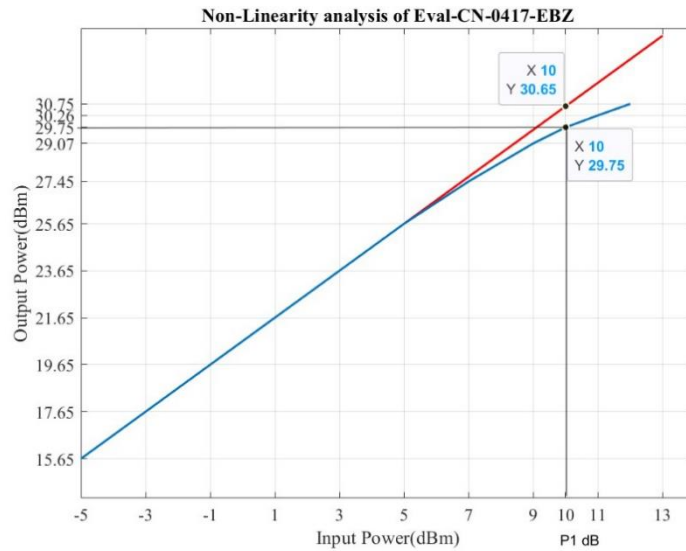


Figure 5.4: Plot of Input VS Output power of the rf power amplifier

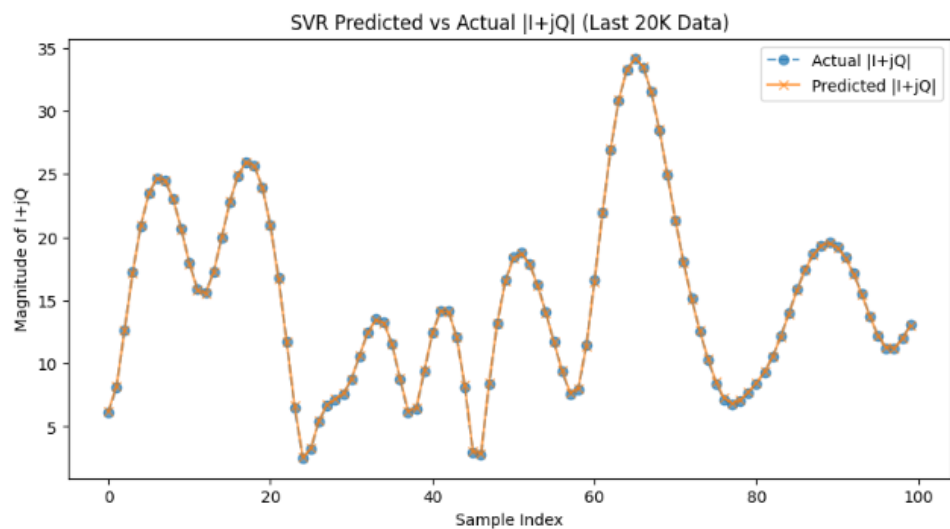
Modeling Using SVR and RBF Model**Forward Model**

Figure 5.5a: Model predicted vs actual graph for I+jQ component

The Support Vector Regression model was optimized using an RBF kernel with parameters $C = 20$, $\epsilon = 0.01$, and $\gamma = 0.05$ for both I and Q components, achieving a accuracy of 97.05% for both components, with a corresponding Normalized Mean Squared Error (NMSE) of -39.21 dB.

Backward model

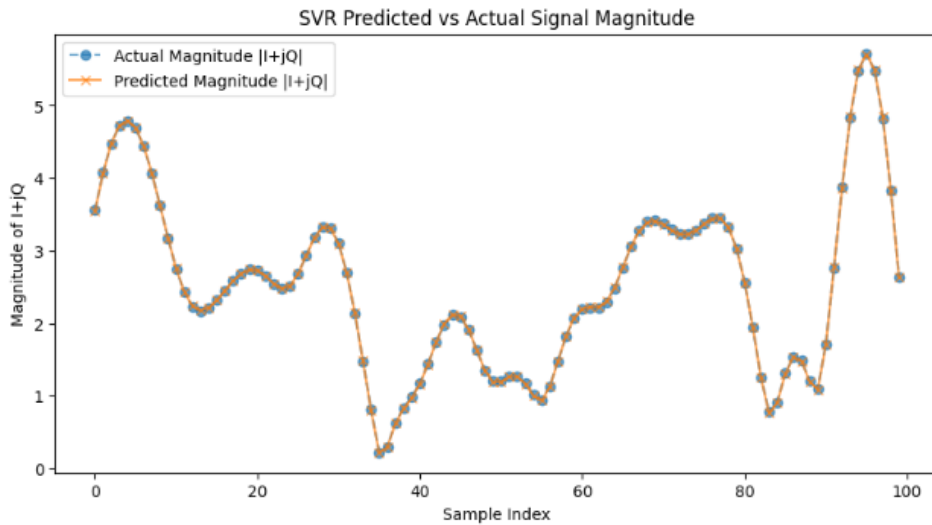


Figure 5.5b: Model predicted graph vs actual graph for I+jQ component

The Support Vector Regression model for the Power Amplifier (PA) was optimized with the RBF kernel, with the parameters of $C = 20$, $\epsilon = 0.01$, and $\gamma = 0.05$ for both the I and Q components. The model achieved a accuracy of 97.61% for both I and Q components, with a Normalized Mean Squared Error (NMSE) of -40.21 dB

Linearization Model

The linearization performance of the DPD system was analyzed at various voltage gain factors ($G = 7.4, 6.2, 6$, and 5.8). The evaluation was conducted using four key plots for each gain: Spectrum, AM/PM, AM/AM in dB, and AM/AM (linear scale)

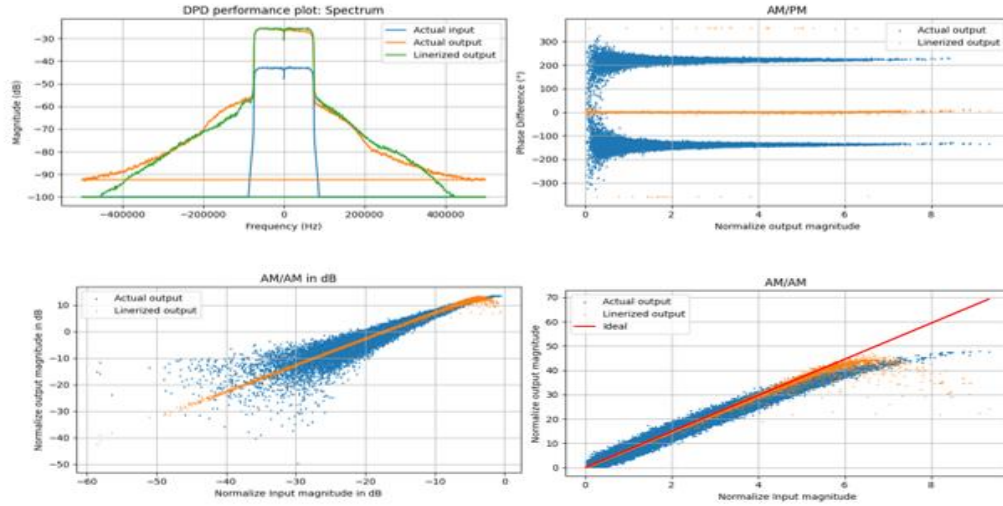


Figure 5.5c: DPD performance plot at gain of 7.4 v/v

From the above figure 5.5c at a gain of 7.4 v/v, the spectrum plot showed noticeable suppression of spectral regrowth, with the linearized output closely following the desired signal's spectrum while significantly reducing the out-of-band emissions compared to the actual PA output. The AM/PM plot demonstrated a substantial reduction in phase distortion after applying DPD, with the linearized output exhibiting a much flatter phase response compared to the PA's output without DPD. In the AM/AM plots, both in linear and dB scales, the linearized output followed the ideal line more accurately than the PA's output. However, minor deviations were still observed, indicating that while the DPD was effective, the system did not achieve perfect linearity at this gain.

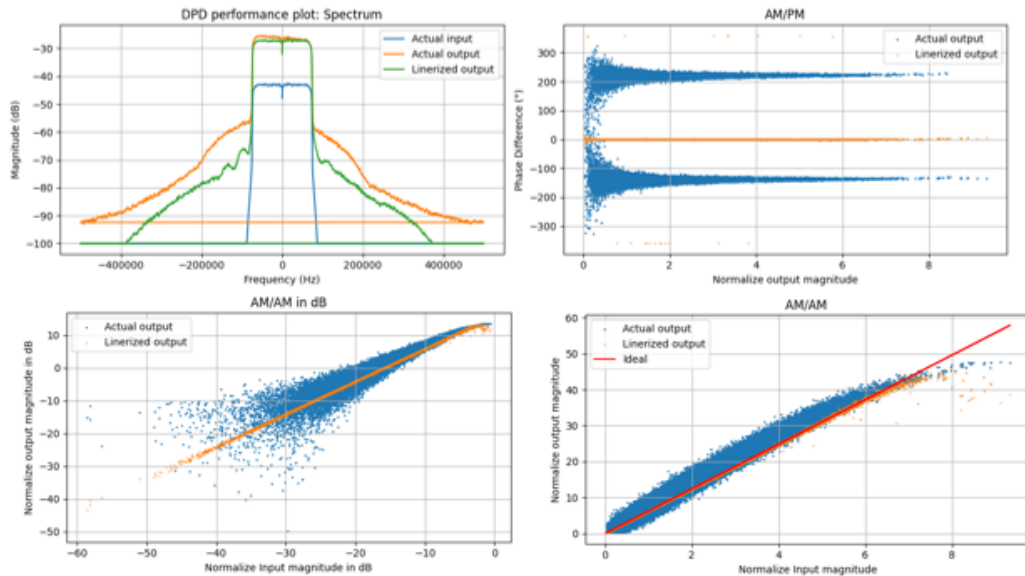


Figure 5.5d: DPD performance plot at gain of 6.2 v/v

From the above figure 5.5d at a slightly lower gain of 6.2 v/v, the performance of the DPD model improved marginally compared to $G = 7.4$. The spectrum plot showed even better suppression of out-of-band components, indicating more effective linearization. The AM/PM curve became flatter, suggesting improved phase correction. The AM/AM plots showed reduced gain compression, with the linearized output tracking the ideal response more consistently across a broader range of input magnitudes. This suggests that the DPD model responded better at this gain, likely due to the PA operating slightly below its most nonlinear region.

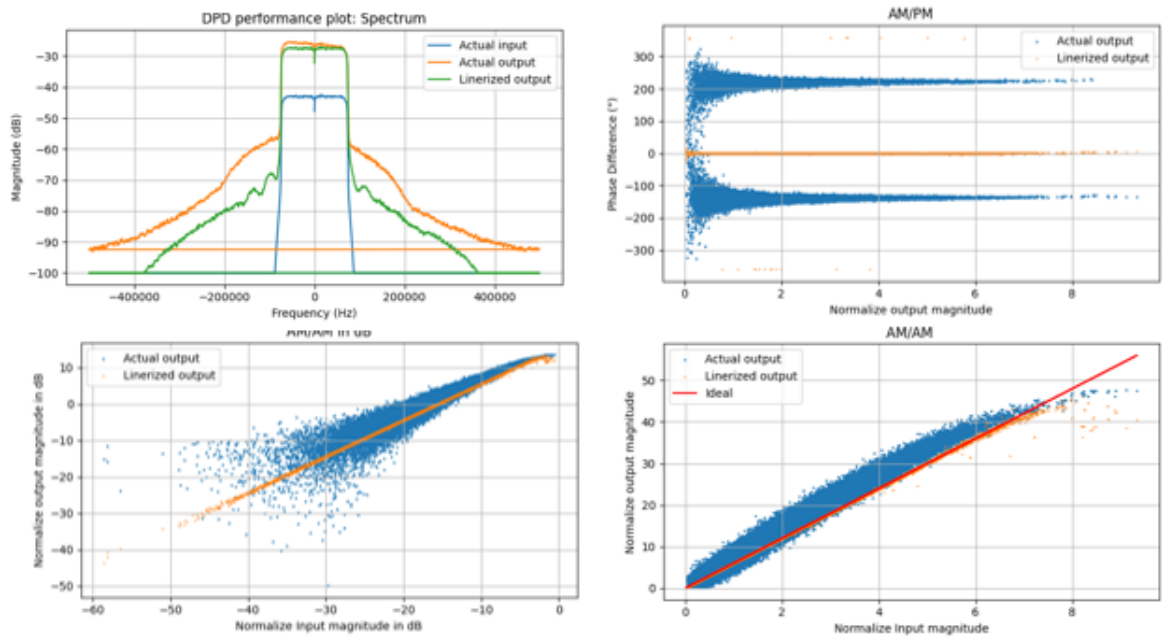


Figure 5.5e: DPD performance plot at gain of 6 v/v

From the above figure 5.5e at $G = 6$ v/v, the linearization performance was further improved. The spectral regrowth was minimal, with the linearized output maintaining the sharpness of the desired signal spectrum and showing significant reduction of sidebands. The AM/PM distortion was almost completely flattened, reflecting minimal phase error after applying DPD. In both AM/AM plots, the linearized output showed excellent alignment with the ideal response, indicating that the DPD model was highly effective at this gain. Overall, this gain setting appeared to be the most optimal among all, achieving the best balance between output power and linearization.

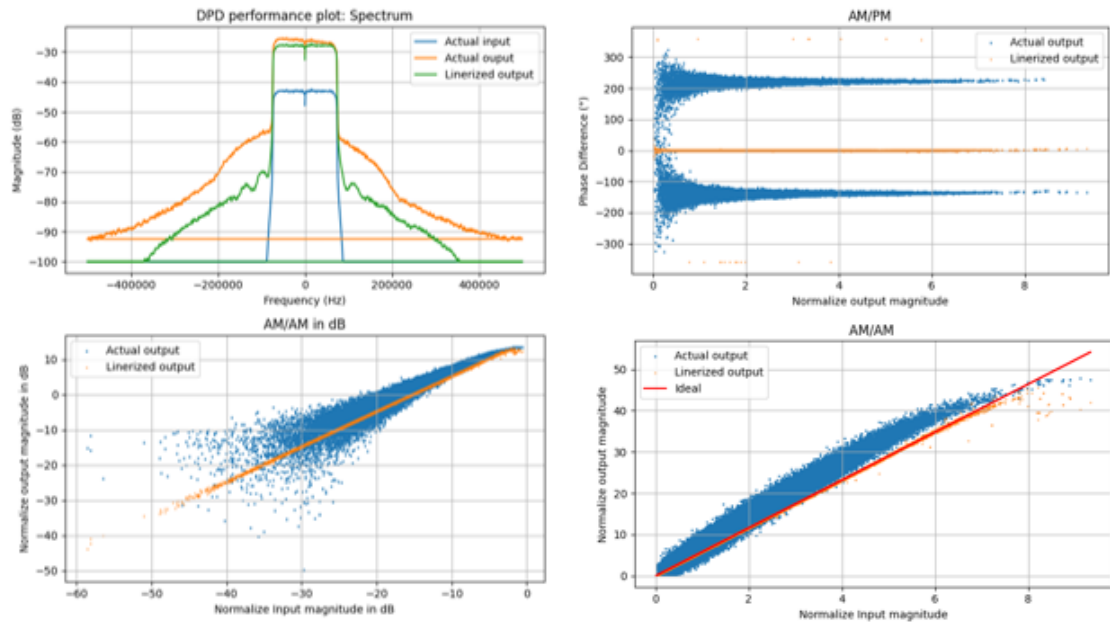


Figure 5.5f: DPD performance plot at gain of 5.8 v/v

From the above fig 5.5f at the lowest gain of 5.8 v/v, the linearization performance remained stable, although slight under-compensation was observed in the AM/AM plots at higher input magnitudes, where the linearized output started to diverge slightly from the ideal line. The spectrum plot continued to show good out-of-band suppression, but the overall linearization performance was slightly less consistent compared to the performance at $G = 6$. The AM/PM plot indicated low phase distortion, but with marginally higher residual errors than at $G = 6$.

Observation Table

Gain (G) (v/v)	Spectrum Regrowth Suppression	AM/AM Linearization Performance	AM/PM Distortion Mitigation	Key Observations
7.4	Moderate suppression observed, with residual sideband components present	Acceptable linearization with observable deviations at higher input magnitudes	Substantial reduction achieved; however, minor residual phase deviations persist	DPD demonstrates improvement, though the PA operates in a higher nonlinear region leading to residual distortions

Gain (G) (v/v)	Spectrum Regrowth Suppression	AM/AM Linearization Performance	AM/PM Distortion Mitigation	Key Observations
6.2	Enhanced suppression with further attenuation of spectral regrowth	Improved linearization, demonstrating closer adherence to the ideal characteristic over a broader input range	Further flattening of phase response with reduced distortion magnitude	Linearization performance improves as the PA operates further from its most nonlinear region
6.0	Superior suppression of out-of-band emissions, indicating minimal spectral leakage	Excellent linearization observed, with minimal gain compression across the entire dynamic range	Phase distortion effectively minimized, with the response approaching ideal conditions	Optimal linearization performance achieved, representing the best compromise between linearity and power efficiency
5.8	Stable suppression maintained; however, minor under-compensation effects evident at higher input levels	Slight deviations from ideal response at elevated input magnitudes, indicating marginal under-compensation	Phase distortions remained low; however, slight residual errors noted	Linearization remains effective; the PA operates predominantly in its linear region, leading to reduced DPD influence

Table 5.1: Comparison table for different Gain

Result

The DPD system improved the linearization performance of the PA across all tested gain values, reducing both amplitude and phase distortions as well as spectral regrowth. At higher gains ($G = 7.4$ and 6.2), DPD effectively reduced distortions, but some residual nonlinearities remained due to the amplifier operating near its saturation region. The best linearization performance was observed at $G = 6$ v/v, where AM/AM and AM/PM distortions were minimized, and spectral purity was maximized, indicating an optimal balance between linearity and efficiency.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

This chapter highlights the key outcomes of the project, emphasizing its overall impact and significance. The future scope explores potential avenues for expanding the work, incorporating new technologies, or enhancing existing features to drive innovation. This chapter aims to offer a comprehensive wrap-up and spark future exploration in the field

6.1 Conclusion

This project successfully demonstrates the design and implementation of a machine learning-based Digital Predistortion (DPD) system for linearizing the nonlinear behaviour of Radio Frequency Power Amplifiers (RFPAs). By leveraging Support Vector Regression (SVR) with a Radial Basis Function (RBF) kernel, both forward and backward models were developed to capture the complex amplitude and phase distortions inherent in real-world amplifier systems. The use of real-time I/Q data, collected from a practical RF setup, allowed for highly accurate modelling and validation of the system's performance under realistic conditions. Through this approach, we achieved significant improvements in linearity metrics, including reduced AM/AM and AM/PM distortion and suppression of spectral regrowth. The linearization model proved effective across multiple gain levels, with optimal performance observed around 6.0 v/v. The results confirm that SVR-based DPD not only provides high accuracy but also offers a scalable solution for modern wireless communication systems.

6.2 Future Scope

Currently, the implemented DPD system is designed for offline training and testing using pre-recorded datasets. While effective for experimentation and evaluation, offline models are not viable for deployment in real-time communication systems. One major future direction is the real-time deployment

of the DPD system using hardware accelerators like Field Programmable Gate Arrays (FPGAs) and Software-Defined Radios (SDRs). FPGAs offer parallel processing capabilities, reconfigurability, and ultra-low latency, making them suitable for real-time digital signal processing (DSP) tasks such as DPD. The SVR model could be translated into a hardware description language (HDL) and mapped onto FPGA logic blocks. This would allow for the predistortion process to be performed on-the-fly as the signal flows through the system, ensuring low delay and high throughput. SDRs, on the other hand, provide flexibility to work across multiple frequency bands and modulation schemes. Implementing the full DPD pipeline—including signal acquisition, SVR inference, and retransmission—on an SDR platform like USRP. While SVR with the RBF kernel is effective in capturing moderate nonlinearity and memory effects, it has limitations when dealing with very complex or high-dimensional signal behaviours. A promising enhancement is the development of hybrid models that combine the strengths of multiple machine learning techniques. For example, SVR can be used as a feature extractor or primary predictor, followed by a neural network layer that refines the output. Alternatively, an ensemble approach could be used where SVR and a neural network run in parallel, and their predictions are averaged or combined through a decision fusion layer. Neural networks, particularly deep learning models like LSTMs or CNNs, are well-suited for learning complex temporal and spatial patterns. When combined with SVR—which offers robustness and mathematical interpretability—such hybrid architectures could result in improved modelling accuracy, better generalization under varying signal conditions, and the potential for online adaptability. Integration of the DPD system into a real-time hardware testbed, enabling thorough validation beyond offline simulations. This testbed would comprise essential components such as Vector Signal Generators (VSGs), Vector Signal Analysers (VSAs), real RF Power Amplifiers (PAs), Software-Defined Radio (SDR) front-ends, and a digital processing unit, which could be an FPGA or an embedded processor. Deploying the system in a hardware environment would allow for comprehensive evaluation under real-world conditions, including noise, signal fading, hardware imperfections, and interference.

REFERENCES

- [1] C. Pérez-Wences, J. R. Loo-Yau, P. Moreno, and J. A. Reynoso-Hernández, "Digital Predistortion of RF Power Amplifier Using the NARX-SVR Model," in *IEEE Microwave and Wireless Technology Letters*, vol. 33, no. 4, pp. 475-478, Apr. 2023, doi: 10.1109/LMWT.2022.3223308.
- [2] C. S. Hemsí and C. M. Panazio, "Sparse Flexible Reduced-Volterra Model for Power Amplifier Digital Pre-Distortion," *IEEE Access*, vol. 10, pp. 121970-121984, 2022
- [3] Sanjika Devi R V, Robin Kalyan, Bindu K. R, Dhanesh G. Kurup, "Optimization of digital predistortion models for RF power amplifiers using a modified differential evolution algorithm," *AEU - International Journal of Electronics and Communications*, Vol. 124, 2020.
- [4] R. V. Sanjika Devi and Dhanesh G. Kurup, "Behavioral modeling of RF Power Amplifiers for designing energy-efficient wireless systems," *Proc. IEEE WiSPNET 2017 Conference*, Chennai, India, March 2017
- [5] Guan and A. Zhu, "Green communications: Digital predistortion for wideband RF power amplifiers," *IEEE Microw. Mag.*, vol. 15, no. 7, pp. 84–99, Nov./Dec. 2014
- [6] Chih-Chung Chang and Chih-Jen Lin, "LIBSVM : a library for support vector machines", *ACM Transactions on Intelligent Systems and Technology*, 2(3), 2011
- [7] F.M Mkadem et al., "Behavioral modeling and digital predistortion of power amplifiers with memory using two hidden layers artificial neural networks," in *IEEE MTT-S Int. Microw. Symp.*, May 2010.
- [8] A. Zhu and J. C. Pedro, "Amplifier distortion evaluation of RF power amplifiers using dynamic deviation reduction based Volterra series," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2007.
- [9] D.R Morgan et al., "A generalized memory polynomial model for digital predistortion of RF power amplifiers," *IEEE Trans. Signal Process.*, vol. 54, no. 10, pp. 3852–3860, Oct. 2006.